Phase 1 Report: τ-Bench Setup and Baseline Evaluation

Team TLDR

February 12, 2026

1 Brief Write-Up

1.1 Project Setup

This phase involved setting up the τ-bench framework on SOL Research Computing Lab resources. We focused on two domains: Airline and Retail, establishing performance baselines for various model sizes and reasoning strategies under simulated multi-turn scenarios.

1.2 Configurations and Models

- **User Agent:** We setup Qwen3-32B as the User Agent which was kept static across all experiments to ensure a fair comparison.
- **Tool Calling Agents:** We utilized the Qwen3 family, specifically the 4B, 8B, 14B, and 32B variants, to analyze the impact of parameter scaling on tool-calling reliability.
- **Inference Environment:** Models were served via vLLM, ensuring consistent temperature and sampling parameters across 5 trials per configuration.
- **Stratagies:** We implemented three distinct strategies - ReAct (Reason + Act), Act, FC (Function Calling) - for the tool-calling agents.

2 Results

The primary metric for evaluation is $pass^k$, which measures the reliability of the agent over k independent trials. The metric is calculated as:

$$pass^k = \mathbb{E}_{\text{task}} \left[\binom{c}{k} \Big/ \binom{n}{k} \right]$$

2.1 Current Progress:

- **Retail Domain:** Evaluated ReAct, Act, and Function Calling (FC) agents using (Qwen3-4B and Qwen3-32B) models completely (115 tasks / trial $\times$ 5 trials), and (Qwen3-8B, Qwen3-14B) models partially.
- **Airline Domain:** Evaluated ReAct agent using all Qwen3 family models (4B, 8B, 14B and 32B) (50 tasks / trial $\times$ 5 trials).

- **Note:** The rows with all values zero, are the partial json files that were stopped midway, due to unavailable SOL sessions, and hence no $pass^k$ scores were calculated, and the rows with zeros in only 1, 2 or all of $pass^3$, $pass^4$, $pass^5$ values indicate that none of the tasks could succeed in more than 4 tasks.

2.2 Performance Summary Table

Table 1: Baseline Results across Retail and Airline Domains

Domain	Model	Method	T0	$pass^1$	T1	$pass^2$	T2	$pass^3$	T3	$pass^4$	T4	$pass^5$
Retail	Qwen3-4B	ReAct	115	0.097	115	0.061	115	0.048	115	0.040	115	0.035
		Act	115	0.101	115	0.063	115	0.044	115	0.033	115	0.026
		FC	115	0.096	115	0.048	115	0.031	115	0.023	115	0.017
	Qwen3-8B	ReAct	115	0.047	115	0.008	24	0.001	0	0.000	0	0.000
		Act	84	0.141	108	0.086	98	0.068	77	0.059	88	0.052
		FC	115	0.143	115	0.067	115	0.043	115	0.031	115	0.026
	Qwen3-14B	ReAct	0	0.000	0	0.000	0	0.000	0	0.000	0	0.000
		Act	115	0.049	40	0.003	0	0.000	0	0.000	0	0.000
		FC	0	0.000	0	0.000	0	0.000	0	0.000	0	0.000
	Qwen3-32B	ReAct	115	0.318	115	0.171	115	0.110	115	0.080	115	0.061
		Act	115	0.197	115	0.090	115	0.056	115	0.042	115	0.035
		FC	115	0.315	115	0.156	115	0.085	115	0.056	115	0.043
Airline	Qwen3-4B	ReAct	50	0.340	50	0.296	50	0.264	50	0.240	50	0.220
		Act	50	0.392	50	0.382	50	0.380	50	0.380	50	0.380
		FC	50	0.328	50	0.266	50	0.236	50	0.216	50	0.200
	Qwen3-8B	ReAct	50	0.288	50	0.206	50	0.152	50	0.112	50	0.080
		Act	50	0.256	50	0.170	50	0.118	50	0.084	50	0.060
		FC	50	0.224	50	0.118	50	0.064	50	0.028	50	0.000
	Qwen3-14B	ReAct	50	0.308	50	0.234	50	0.194	50	0.172	50	0.160
		Act	50	0.280	50	0.204	50	0.166	50	0.140	50	0.120
		FC	50	0.252	50	0.120	50	0.068	50	0.048	50	0.040
	Qwen3-32B	ReAct	50	0.300	50	0.174	50	0.118	50	0.084	50	0.060
		Act	50	0.316	50	0.198	50	0.132	50	0.088	50	0.060
		FC	50	0.260	50	0.114	50	0.056	50	0.024	50	0.000

3 Individual Contributions

The project workload was distributed among the five team members as follows:

1. **Vedang:** Executed (Qwen3-32B + Qwen3-32B) User-Agent pair, for the Airline domain. Due to the high resource demands of the 32B model, tasks were split across 2 GPUs with adjusted context lengths compatible with vLLM. Utilized 4-bit quantization via ‘bitsandbytes‘ to accommodate the varying parameters of SOL requests.

2. **Harshith:** Executed (Qwen3-32B + Qwen3-4B) User-Agent pair on both Retail and Airline domains. Utilized 4-bit quantization via ‘bitsandbytes‘ to accommodate the varying parameters of SOL requests.

3. **Shravan:** Executed (Qwen3-32B + Qwen3-8B) User-Agent pair on both Retail and Airline domains. Utilized 4-bit quantization via 'bitsandbytes' to accommodate the varying parameters of SOL requests.

4. **Vidya:** Executed (Qwen3-32B + Qwen3-32B) User-Agent pair for the Retail domain. Due to the high resource demands of the 32B model, tasks were split across 2 GPUs with adjusted context lengths compatible with vLLM. Utilized 4-bit quantization via 'bitsandbytes' to accommodate the varying parameters of SOL requests.

5. **Gouri:** Executed (Qwen3-32B + Qwen3-14B) User-Agent pair on both Retail and Airline domains. Utilized 4-bit quantization via 'bitsandbytes' to accommodate the varying parameters of SOL requests.

4 Challenges Faced

4.1 Environment & Build Complexity

- Initial attempts to use mamba for fast creation of virtue environments failed due to conflicts with Intel's custom PyTorch extensions (IPEX). We had to move back to a standard conda setup but faced broken pip dependencies ('pkg resources' warnings).
- The standard habana main branch of vllm-fork was missing critical C++ source code ('csrc / hpu'), requiring a pivot to the 'gaudi_1_16_0 release' tag and manual git-based integration of the 'vllm-hpu-extension' repo.

4.2 Hardware & Memory Constraints

- Scaling from 14B to 32B models revealed strict hardware limits. A single Qwen3-32B model (FP16) required 65GB of VRAM, pushing the limits of the Gaudi accelerator (around 96GB HBM) when running dual servers (Agent + User Simulator).
- Initial attempts to run both models on a single card with invalid 'gpu-memory-utilization' errors caused immediate Out-Of-Memory (OOM) crashes. We resolved this by implementing the Agent on Device 0 and User on Device 1, to increase the utilization per card.
- We also had to request short sessions on SOL to make sure our requests are not delayed and drastically reduce our FairShare scores, which also weren't sufficient to continuously monitor the code runs.
- The initialization time for the 32B model on HPU exceeded standard timeout thresholds, causing '**litellm.NotFoundError**' exceptions. We tuned the startup delay to around 2 minutes, and implemented checkpointing as well, so that if the SOL session abruptly cancels due to time limit, the code can safely be continued from where it left off.

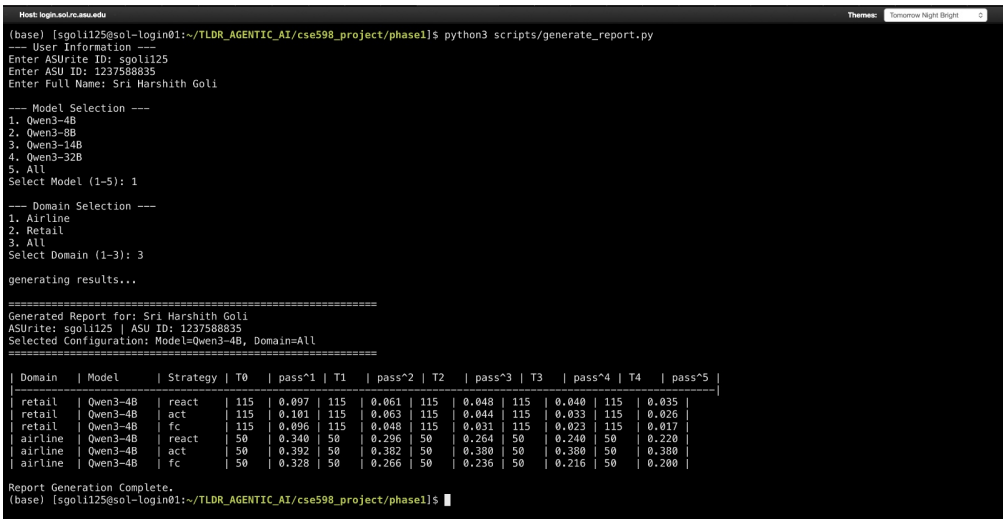
4.3 Context Window & Dynamic Shapes

- The default context length of 32,768 tokens was not sufficient for the 32B model, consuming excessive KV cache memory. We were forced to dynamically update the context-window to ensure stability of the model and not crashing repeatedly at particular tasks.
- The Gaudi Torch compilation backend crashed when handling dynamic input shapes in 'tau-bench' workloads. We had to use git-based installations for vLLM to make it compatible with Gaudi envs.

4.4 Additional Resources

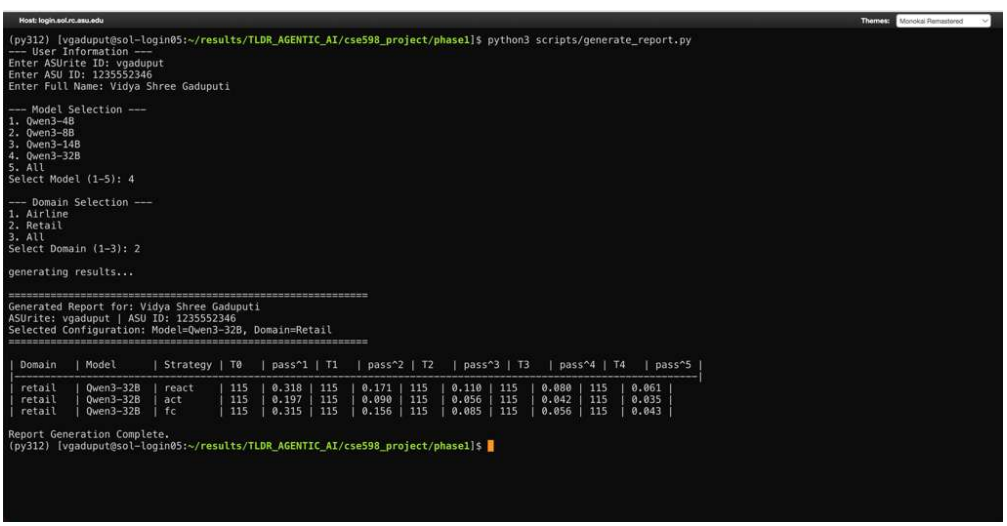
All experimental files, datasets, and detailed logs are available in our Google Drive repository: Click here for dropbox link

4.5 Screenshots of Results



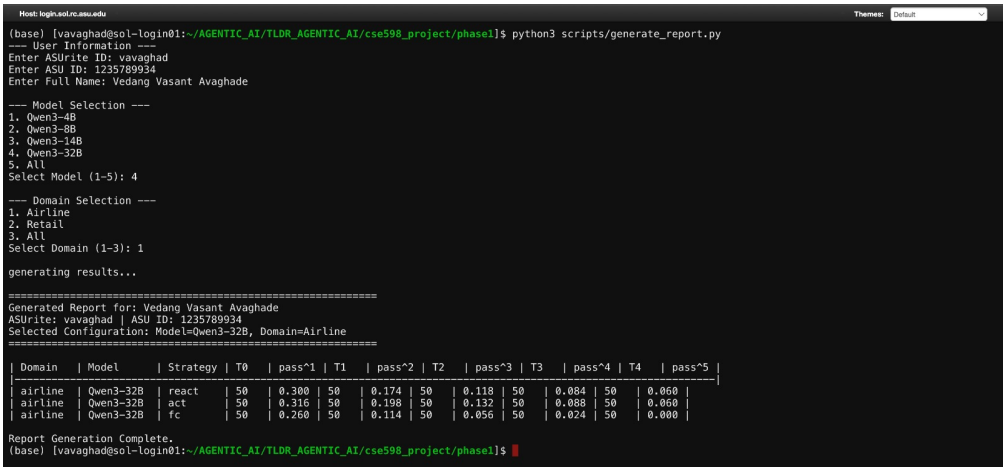
```
Host: login.sol.rc.asu.edu                                                    Themes: Tomorrow Night Bright

(base) [sgoli125@sol-login01:~/TLDR_AGENTIC_AI/cse598_project/phase1]$ python3 scripts/generate_report.py
--- User Information ---
Enter ASUrite ID: sgoli125
Enter ASU ID: 1237588835
Enter Full Name: Sri Harshith Goli

--- Model Selection ---
1. Qwen3-4B
2. Qwen3-8B
3. Qwen3-14B
4. Qwen3-32B
5. All
Select Model (1-5): 1

--- Domain Selection ---
1. Airline
2. Retail
3. All
Select Domain (1-3): 3

generating results...

============================================================
Generated Report for: Sri Harshith Goli
ASUrite: sgoli125 | ASU ID: 1237588835
Selected Configuration: Model=Qwen3-4B, Domain=All
============================================================

| Domain   | Model      | Strategy | T0   | pass^1 | T1   | pass^2 | T2   | pass^3 | T3   | pass^4 | T4   | pass^5 |
|----------|------------|----------|------|--------|------|--------|------|--------|------|--------|------|--------|
| retail   | Qwen3-4B   | react    | 115  | 0.097  | 115  | 0.061  | 115  | 0.048  | 115  | 0.040  | 115  | 0.035  |
| retail   | Qwen3-4B   | act      | 115  | 0.101  | 115  | 0.063  | 115  | 0.044  | 115  | 0.033  | 115  | 0.026  |
| retail   | Qwen3-4B   | fc       | 115  | 0.096  | 115  | 0.048  | 115  | 0.031  | 115  | 0.023  | 115  | 0.017  |
| airline  | Qwen3-4B   | react    | 50   | 0.340  | 50   | 0.296  | 50   | 0.264  | 50   | 0.240  | 50   | 0.220  |
| airline  | Qwen3-4B   | act      | 50   | 0.392  | 50   | 0.382  | 50   | 0.380  | 50   | 0.380  | 50   | 0.380  |
| airline  | Qwen3-4B   | fc       | 50   | 0.328  | 50   | 0.266  | 50   | 0.236  | 50   | 0.216  | 50   | 0.200  |

Report Generation Complete.
(base) [sgoli125@sol-login01:~/TLDR_AGENTIC_AI/cse598_project/phase1]$
```

Figure 1: Qwen3-4B (Retail + Airline) domain results



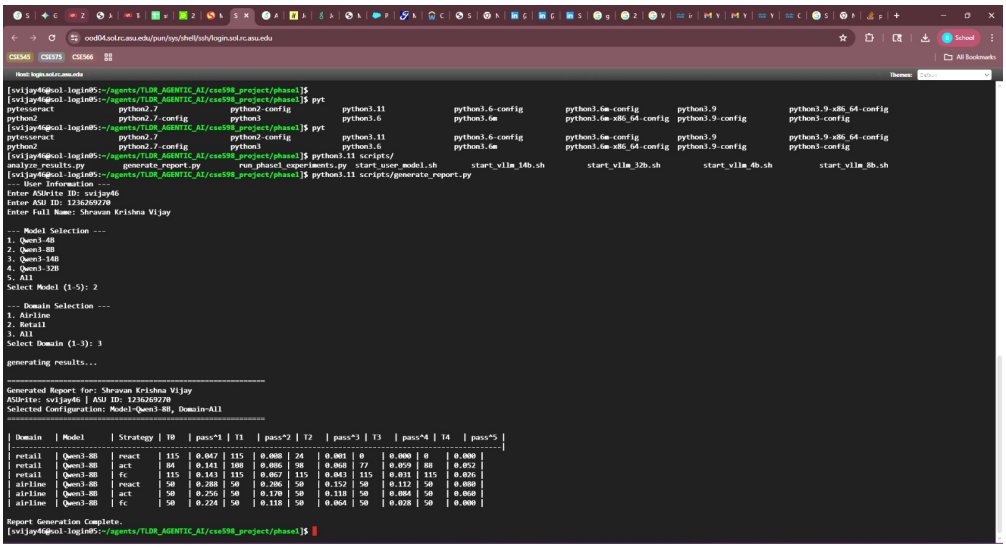
```
Host: login.sol.rc.asu.edu                                                    Themes: Monokai Remastered

(py312) [vgaduput@sol-login05:~/results/TLDR_AGENTIC_AI/cse598_project/phase1]$ python3 scripts/generate_report.py
--- User Information ---
Enter ASUrite ID: vgaduput
Enter ASU ID: 1235552346
Enter Full Name: Vidya Shree Gaduputi

--- Model Selection ---
1. Qwen3-4B
2. Qwen3-8B
3. Qwen3-14B
4. Qwen3-32B
5. All
Select Model (1-5): 4

--- Domain Selection ---
1. Airline
2. Retail
3. All
Select Domain (1-3): 2

generating results...

============================================================
Generated Report for: Vidya Shree Gaduputi
ASUrite: vgaduput | ASU ID: 1235552346
Selected Configuration: Model=Qwen3-32B, Domain=Retail
============================================================

| Domain   | Model      | Strategy | T0   | pass^1 | T1   | pass^2 | T2   | pass^3 | T3   | pass^4 | T4   | pass^5 |
|----------|------------|----------|------|--------|------|--------|------|--------|------|--------|------|--------|
| retail   | Qwen3-32B  | react    | 115  | 0.318  | 115  | 0.171  | 115  | 0.110  | 115  | 0.080  | 115  | 0.061  |
| retail   | Qwen3-32B  | act      | 115  | 0.197  | 115  | 0.090  | 115  | 0.056  | 115  | 0.042  | 115  | 0.035  |
| retail   | Qwen3-32B  | fc       | 115  | 0.315  | 115  | 0.156  | 115  | 0.085  | 115  | 0.056  | 115  | 0.043  |

Report Generation Complete.
(py312) [vgaduput@sol-login05:~/results/TLDR_AGENTIC_AI/cse598_project/phase1]$
```

Figure 2: Qwen3-32B Retail domain results



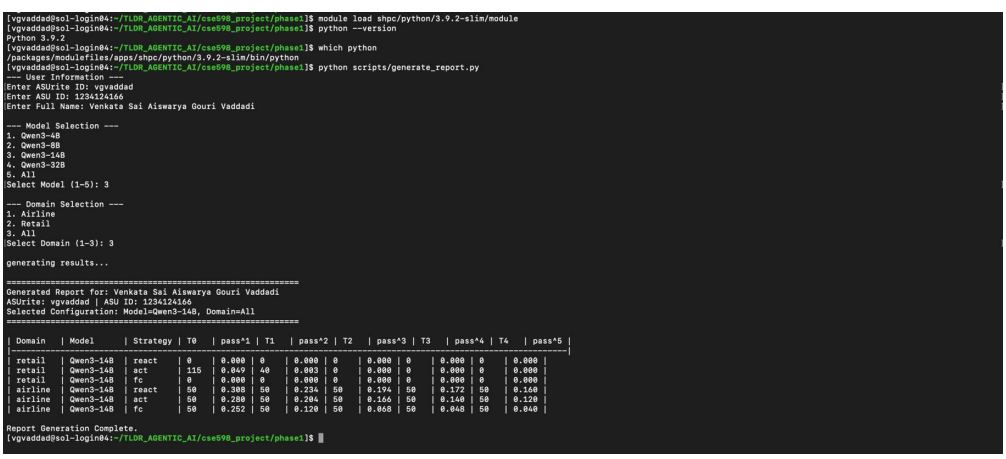
```
Host: login.sol.rc.asu.edu                                                                                    Themes: Default

(base) [vavaghad@sol-login01:~/AGENTIC_AI/TLDR_AGENTIC_AI/cse598_project/phase1]$ python3 scripts/generate_report.py
--- User Information ---
Enter ASUrite ID: vavaghad
Enter ASU ID: 1235789934
Enter Full Name: Vedang Vasant Avaghade

--- Model Selection ---
1. Qwen3-4B
2. Qwen3-8B
3. Qwen3-14B
4. Qwen3-32B
5. All
Select Model (1-5): 4

--- Domain Selection ---
1. Airline
2. Retail
3. All
Select Domain (1-3): 1

generating results...

============================================================
Generated Report for: Vedang Vasant Avaghade
ASUrite: vavaghad | ASU ID: 1235789934
Selected Configuration: Model=Qwen3-32B, Domain=Airline
============================================================

| Domain   | Model      | Strategy | T0   | pass^1 | T1   | pass^2 | T2   | pass^3 | T3   | pass^4 | T4   | pass^5 |
|----------|------------|----------|------|--------|------|--------|------|--------|------|--------|------|--------|
| airline  | Qwen3-32B  | react    | 50   | 0.300  | 50   | 0.174  | 50   | 0.118  | 50   | 0.084  | 50   | 0.060  |
| airline  | Qwen3-32B  | act      | 50   | 0.316  | 50   | 0.198  | 50   | 0.132  | 50   | 0.088  | 50   | 0.060  |
| airline  | Qwen3-32B  | fc       | 50   | 0.260  | 50   | 0.114  | 50   | 0.056  | 50   | 0.024  | 50   | 0.000  |

Report Generation Complete.
(base) [vavaghad@sol-login01:~/AGENTIC_AI/TLDR_AGENTIC_AI/cse598_project/phase1]$
```

Figure 3: Qwen3-32B Airline domain results



```
[svijay46@sol-login05:~/agents/TLDR_AGENTIC_AI/cse598_project/phase1]$
[svijay46@sol-login05:~/agents/TLDR_AGENTIC_AI/cse598_project/phase1]$ pyt
pytesseract          python2.7            python2-config       python3.11           python3.6-config     python3.6m-config    python3.9            python3.9-x86_64-config
python2              python2.7-config     python3              python3.6            python3.6m           python3.6m-x86_64-config  python3.9-config     python3-config
[svijay46@sol-login05:~/agents/TLDR_AGENTIC_AI/cse598_project/phase1]$ pyt
pytesseract          python2.7            python2-config       python3.11           python3.6-config     python3.6m-config    python3.9            python3.9-x86_64-config
python2              python2.7-config     python3              python3.6            python3.6m           python3.6m-x86_64-config  python3.9-config     python3-config
[svijay46@sol-login05:~/agents/TLDR_AGENTIC_AI/cse598_project/phase1]$ python3.11 scripts/
analyze_results.py   generate_report.py   run_phase1_experiments.py  start_user_model.sh   start_vllm_14b.sh   start_vllm_32b.sh   start_vllm_4b.sh   start_vllm_8b.sh
[svijay46@sol-login05:~/agents/TLDR_AGENTIC_AI/cse598_project/phase1]$ python3.11 scripts/generate_report.py
--- User Information ---
Enter ASUrite ID: svijay46
Enter ASU ID: 1236269270
Enter Full Name: Shravan Krishna Vijay

--- Model Selection ---
1. Qwen3-4B
2. Qwen3-8B
3. Qwen3-14B
4. Qwen3-32B
5. All
Select Model (1-5): 2

--- Domain Selection ---
1. Airline
2. Retail
3. All
Select Domain (1-3): 3

generating results...

============================================================
Generated Report for: Shravan Krishna Vijay
ASUrite: svijay46 | ASU ID: 1236269270
Selected Configuration: Model=Qwen3-8B, Domain=All
============================================================

| Domain   | Model      | Strategy | T0   | pass^1 | T1   | pass^2 | T2   | pass^3 | T3   | pass^4 | T4   | pass^5 |
|----------|------------|----------|------|--------|------|--------|------|--------|------|--------|------|--------|
| retail   | Qwen3-8B   | react    | 115  | 0.047  | 115  | 0.008  | 24   | 0.003  | 0    | 0.000  | 0    | 0.000  |
| retail   | Qwen3-8B   | act      | 84   | 0.141  | 108  | 0.086  | 98   | 0.068  | 77   | 0.059  | 88   | 0.052  |
| retail   | Qwen3-8B   | fc       | 115  | 0.143  | 115  | 0.067  | 115  | 0.043  | 115  | 0.031  | 115  | 0.026  |
| airline  | Qwen3-8B   | react    | 50   | 0.288  | 50   | 0.206  | 50   | 0.152  | 50   | 0.112  | 50   | 0.080  |
| airline  | Qwen3-8B   | act      | 50   | 0.256  | 50   | 0.170  | 50   | 0.118  | 50   | 0.084  | 50   | 0.060  |
| airline  | Qwen3-8B   | fc       | 50   | 0.224  | 50   | 0.118  | 50   | 0.064  | 50   | 0.028  | 50   | 0.000  |

Report Generation Complete.
[svijay46@sol-login05:~/agents/TLDR_AGENTIC_AI/cse598_project/phase1]$
```

Figure 4: Qwen3-8B (Retail + Airline) domain results



```
[vgvaddad@sol-login04:~/TLDR_AGENTIC_AI/cse598_project/phase1]$ module load shpc/python/3.9.2-slim/module
[vgvaddad@sol-login04:~/TLDR_AGENTIC_AI/cse598_project/phase1]$ python --version
Python 3.9.2
[vgvaddad@sol-login04:~/TLDR_AGENTIC_AI/cse598_project/phase1]$ which python
/packages/modulefiles/apps/shpc/python/3.9.2-slim/bin/python
[vgvaddad@sol-login04:~/TLDR_AGENTIC_AI/cse598_project/phase1]$ python scripts/generate_report.py
--- User Information ---
Enter ASUrite ID: vgvaddad
Enter ASU ID: 1234124166
Enter Full Name: Venkata Sai Aiswarya Gouri Vaddadi

--- Model Selection ---
1. Qwen3-4B
2. Qwen3-8B
3. Qwen3-14B
4. Qwen3-32B
5. All
Select Model (1-5): 3

--- Domain Selection ---
1. Airline
2. Retail
3. All
Select Domain (1-3): 3

generating results...

============================================================
Generated Report for: Venkata Sai Aiswarya Gouri Vaddadi
ASUrite: vgvaddad | ASU ID: 1234124166
Selected Configuration: Model=Qwen3-14B, Domain=All
============================================================

| Domain   | Model      | Strategy | T0   | pass^1 | T1   | pass^2 | T2   | pass^3 | T3   | pass^4 | T4   | pass^5 |
|----------|------------|----------|------|--------|------|--------|------|--------|------|--------|------|--------|
| retail   | Qwen3-14B  | react    | 0    | 0.000  | 0    | 0.000  | 0    | 0.000  | 0    | 0.000  | 0    | 0.000  |
| retail   | Qwen3-14B  | act      | 115  | 0.049  | 40   | 0.003  | 0    | 0.000  | 0    | 0.000  | 0    | 0.000  |
| retail   | Qwen3-14B  | fc       | 0    | 0.000  | 0    | 0.000  | 0    | 0.000  | 0    | 0.000  | 0    | 0.000  |
| airline  | Qwen3-14B  | react    | 50   | 0.308  | 50   | 0.234  | 50   | 0.194  | 50   | 0.172  | 50   | 0.140  |
| airline  | Qwen3-14B  | act      | 50   | 0.280  | 50   | 0.204  | 50   | 0.166  | 50   | 0.140  | 50   | 0.120  |
| airline  | Qwen3-14B  | fc       | 50   | 0.252  | 50   | 0.120  | 50   | 0.068  | 50   | 0.048  | 50   | 0.040  |

Report Generation Complete.
[vgvaddad@sol-login04:~/TLDR_AGENTIC_AI/cse598_project/phase1]$
```

Figure 5: Qwen3-14B (Retail + Airline) domain results